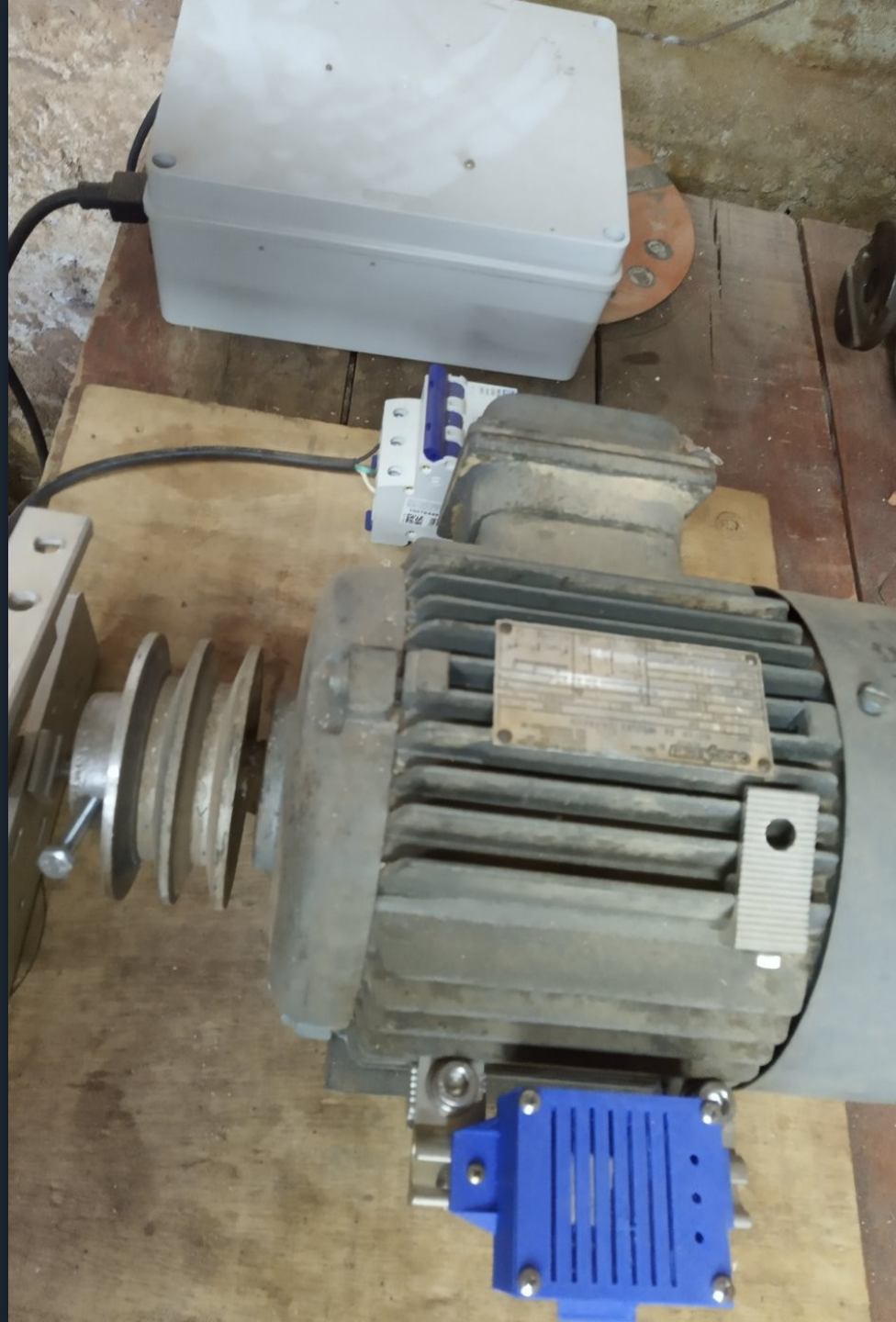
Open-Source Predictive Maintenance System for AC Induction Motors Using LSTM Neural Networks and Low-Cost Embedded Hardware

Proof-of-concept laboratory validation

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Why low-cost predictive maintenance

43–46 %

of global electricity is used by electric motor-driven systems [1]

30–50 %

less unplanned downtime with predictive maintenance [3]

10–25 %

lower maintenance costs than reactive strategies [3]

But SMEs in developing regions still run corrective or time-based maintenance

- Recurring subscription fees
- Proprietary data silos
- Limited access to historical data

Goal: an end-to-end, reproducible system built only from commodity hardware and open-source software.

Contributions

Open hardware node

ESP8266 + MPU-6050 + LM35, bill of materials below USD 15 per node, locally sourced

Signal chain + ISO reference

Acceleration-to-velocity processing, with ISO 20816-1 zones as reference thresholds

Wireless gateway

Raspberry Pi gateway with its own access point: no cabling changes on the plant floor

Full public release

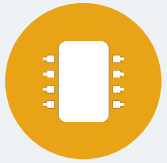
Schematics, PCB, firmware, gateway code and documentation on GitHub

LSTM forecasting

Hyperparameter search evaluated with a similarity score and point-wise error metrics

github.com/Juandeleon-utec/openpdm-ac-motors

Three-tier architecture



1 Sensing node

- ESP8266 + MPU-6050 + LM35
- 1 kHz sampling, on-node features
- One record per second

Wi-Fi →



2 Plant-floor gateway

- Wi-Fi router: access point + DHCP
- Raspberry Pi 3B: Apache, PHP
- MariaDB local storage

LAN →

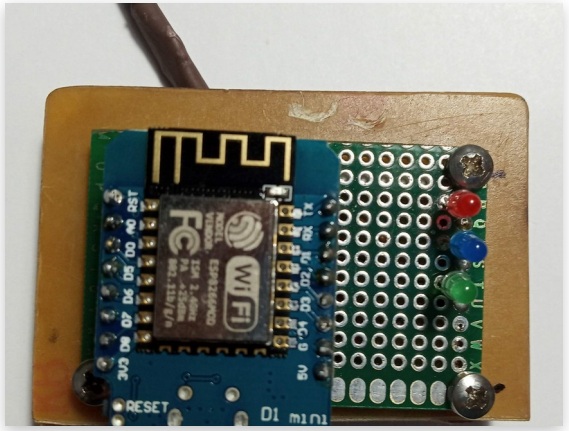


3 Analytics server

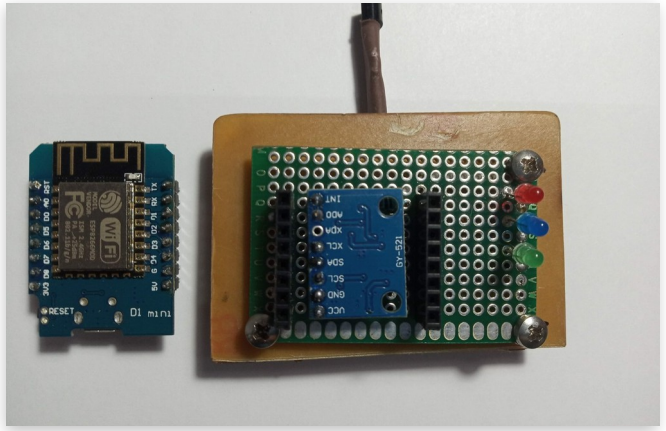
- LSTM training and forecasting
- Python, TensorFlow / Keras
- Grafana / InfluxDB dashboards

Each node is identified by its MAC address (plug-and-play). All software is open source: Arduino, MariaDB, TensorFlow/Keras, KiCad, Grafana/InfluxDB, Raspberry Pi OS.

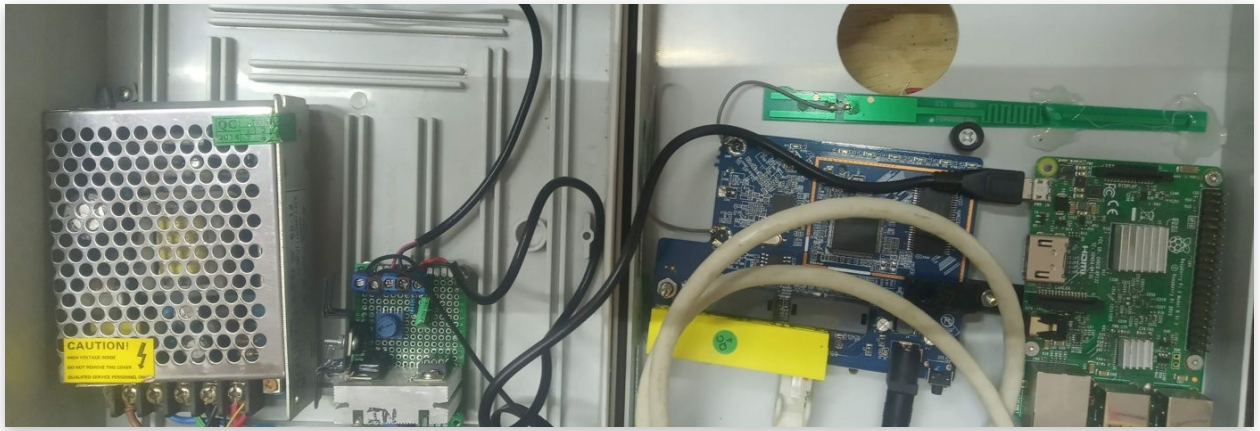
Sensing node and gateway hardware



(a) Assembled node



(b) GY-521 (MPU-6050) under the D1 Mini



(c) Gateway: power supply, Wi-Fi router, Raspberry Pi 3B

Wemos D1 Mini (ESP8266)

80 MHz, Wi-Fi 802.11 b/g/n, MAC-based ID

MPU-6050 on GY-521

Triaxial MEMS, ±2 g, I²C at 400 kHz

LM35 temperature probe

10 mV/°C; monitored, not a model input

3D-printed PLA enclosure

Bolted to the frame through an aluminum bracket

< USD 15 bill of materials per node

On-node signal processing

1 — Sample
fs = 1 kHz
N = 500 in 0.5 s

→

2 — High-pass
fc = 10 Hz
removes gravity and drift

→

3 — Integrate
trapezoidal rule
Δt = 1 ms → v_RMS

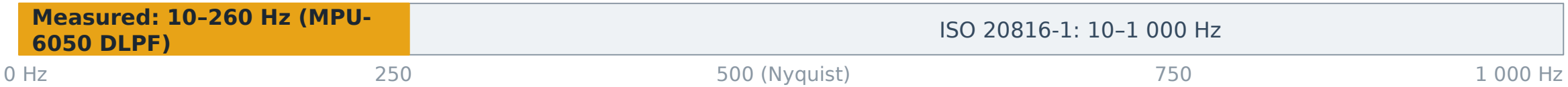
→

4 — Features
$\sigma^2 a$ per axis
X axis → LSTM input

→

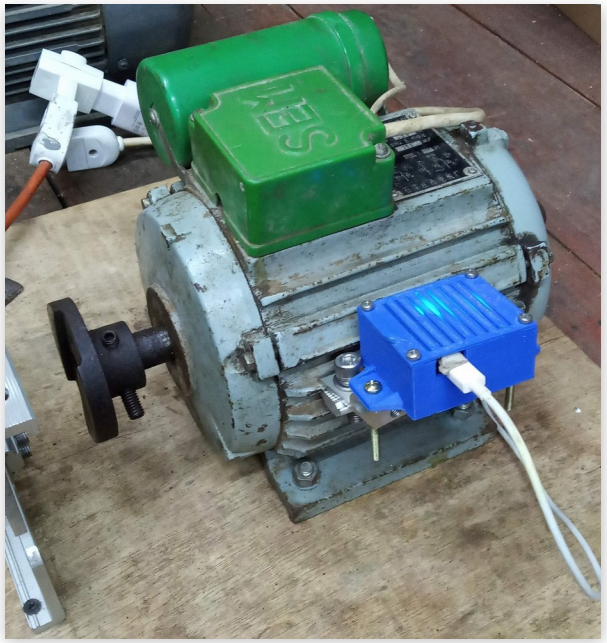
5 — Transmit
1 record / s
Wi-Fi → MariaDB

Measurement band vs. ISO 20816-1



- Covers up to the 5th harmonic of the single-phase motor (47.3 Hz) and the 10th of the three-phase motor (24.2 Hz): imbalance, misalignment, looseness.
- Integration weights each component by 1/(2πf), so the missing 260–1 000 Hz band adds little to v_RMS here, but early bearing defects are not captured.
- **ISO zones are used as reference alarm thresholds, not as a standard-compliant measurement.**

Experimental setup



Single-phase
550 W · 2 840 r/min · 220 V



Three-phase
1.1 kW · 1 450 r/min · 380 V

One sensing node, mounted on one motor at a time (2020).

Protocol 1 — Static imbalance

Four calibrated masses (13.39–43.37 g) on the shaft at r = 25 mm; five 9-min sessions per level.

+35.6 % smallest mass **+280.3 %** largest

X-axis std. deviation vs. baseline, consistent on both motors

Protocol 2 — Progressive loading

- Lever with counterweight (2.75 kg in Exp. 1, 3.55 kg in Exp. 2) and a 5 L reservoir emptying by drip
- Radial load increases without stopping the motor
- 4 500 records per run (1 per second, ≈ 75 min)

LSTM model and evaluation

Chronological split of each run (4 500 records)



Hyperparameters selected on validation; test used only once. Z-score normalization.



4 385 parameters
RMSprop · MSE loss · batch 16 · 800 epochs · Python, TensorFlow/Keras

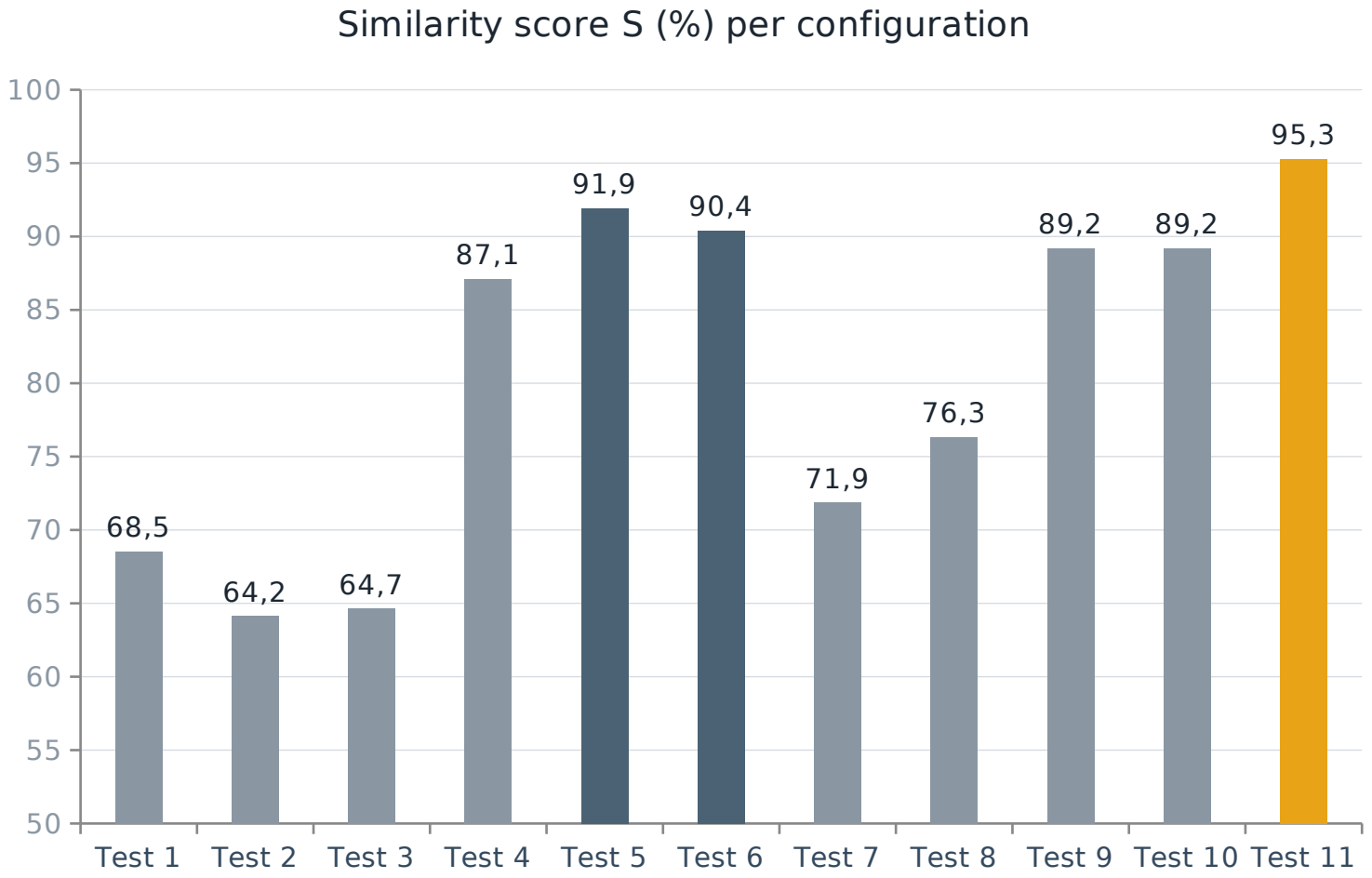
Similarity score S

$$S = 100 \cdot (1 - \tfrac{1}{3} \Sigma\, |\theta_f - \theta| \,/\, |\theta|) \;\; \%$$

$\theta \in$ {mean, variance, std. deviation} of the observed series; θ_f the same statistic of the forecast.

- Measures distribution similarity, not trajectory agreement
- Complemented with RMSE, MAE, R^2 and r

Partial hyperparameter search: 11 configurations



Test 11 (selected)

95.30 %

1 layer · 32 units · batch 16 · 800 epochs · 744 s training

- Two-layer models and 1 600 epochs scored lower
- Grey: other single runs; steel: next best (Tests 5, 6)
- One run per configuration: small gaps between the top three are within run-to-run noise

Point-wise metrics on held-out data

Model	Subset	RMSE	MAE	R^2	r
Test 6	A	0.785	0.601	**−0.891**	0.156
Test 5	A	0.456	0.366	0.287	0.771
Test 11	**A**	**0.404**	**0.323**	**0.457**	**0.819**
Test 11	B	0.574	0.430	**−0.076**	0.589

Normalized units. Subset B is three times the size of subset A; exact composition could not be reconstructed from the 2020 logs.

Point-wise metrics confirm the ranking given by S on subset A: Test 11 > Test 5 > Test 6.



S can mislead

Test 6: S = 90.41 % but $R^2 = -0.891$. Matching statistics do not mean matching trajectories.

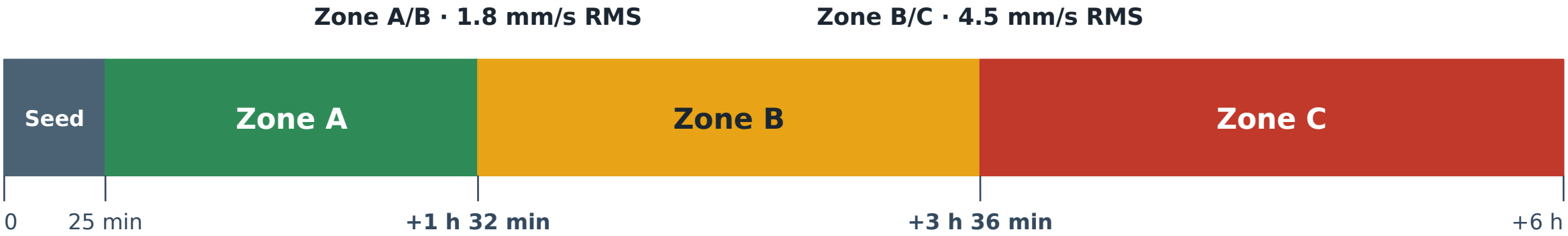


Trend yes, level no

Test 11 tracks the trend on A (r = 0.82) but not on the larger subset B ($R^2 \approx 0$): a chronological split of a monotonic signal forces extrapolation.

Long-range autoregressive forecast

Independent run (Experiment 2, counterweight 3.55 kg): 1 500 real seed records, then 21 500 self-fed predictions (≈ 6 h)



Indicative, not calibrated

- Times come from a heuristic indicator (mean + variance of the forecast) compared against velocity thresholds; the exact mapping could not be reconstructed.
- Both crossings fall beyond the 75 min observed record: they are projections. A consistent mapping, or forecasting v_RMS directly, is future work.

From the 2020 proof of concept to the master's thesis

2020 limitation	Master's thesis (ongoing)
Band 10–260 Hz (MPU-6050)	ESP32-S3 + ADXL355, full 10–1 000 Hz band
Sensor mounted near the feet	Mounting on the bearing housings (ISO 20816-1)
Heuristic mapping to mm/s	Forecast v_RMS directly, validated against a reference vibrometer
No baselines, one run per config	Persistence, linear and ARIMA baselines; repeated seeds
Laboratory, two motors	TRL 7 deployment in an agro-industrial plant (> 100 motors)
Original datasets lost	Data, code and hardware versioned in an open repository from day one

Conclusions

1. A complete, open-source monitoring chain can be built for less than USD 15 per node
2. A compact LSTM (4 385 parameters) captures the degradation trend, but not the level beyond the training range
3. A similarity score alone can mislead: point-wise metrics are essential
4. The 2020 limitations define the roadmap of the master's thesis

Code, hardware and documentation

github.com/Juandeleon-utec/openpdm-ac-motors

Thank you · Questions?

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